

# yadiggg: Strategic Decision Map and Roadmap

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## 1. Operating System & Firmware Resolution

To resolve the core firmware architecture confusion between a Real-Time Operating System (RTOS) and Embedded Linux:

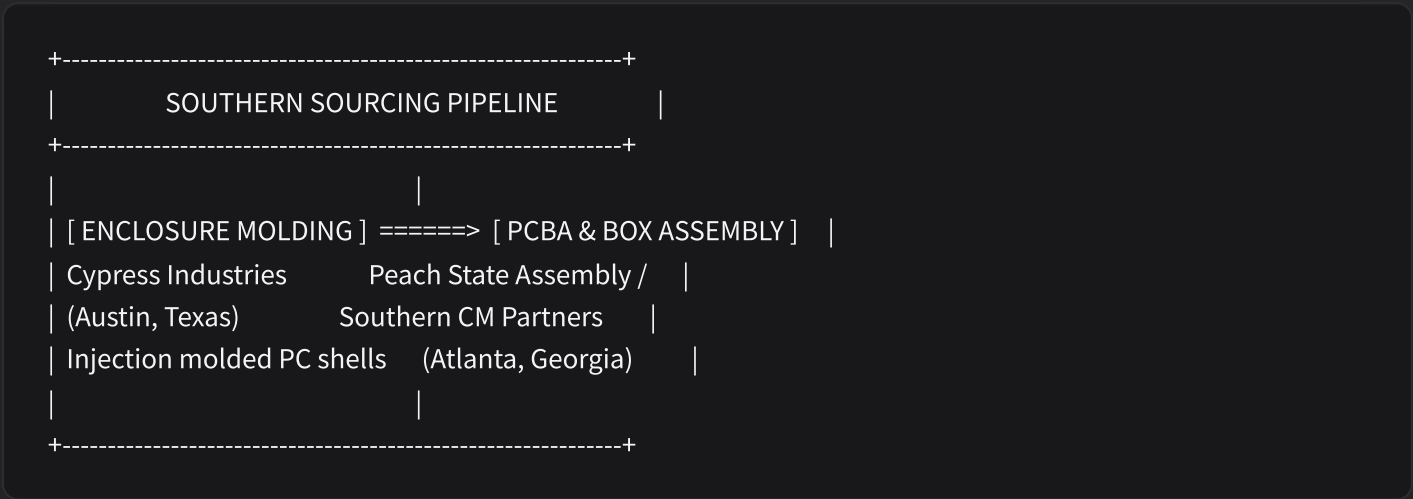
### 1.1 Firmware Stack Commitment: Embedded Linux

We formally rule out the use of a standalone RTOS (like FreeRTOS or Zephyr) as the primary operating system. The computational requirements of the yadiggg intelligence engine require virtual memory space, robust multi-threading, and standard Unix file-system drivers.

- **The OS Choice: Embedded Linux** (built via the **Yocto Project**, utilizing a highly stripped-down, customized Linux Kernel).
- **Why Linux?:**
  1. **SQLite:** Running a local database containing 1,000,000 index files with fast index access (FTS5) requires the robust virtual-memory caching of the Linux VFS (Virtual File System).
  2. **Tesseract-lite:** The OCR text localization and classification libraries are written in C/C++ and depend on standard Unix POSIX libraries that are native to Linux.
  3. **USB-C Mass Storage:** Interfacing with the desktop app as a standard USB Mass Storage device or via MTP (Media Transfer Protocol) to sync database files is natively supported by the Linux USB Gadget driver stack.
- **Standby/Boot Optimization:** To achieve a "disappearing device" feel, the Linux kernel will be stripped of all unnecessary drivers, systemd services, and network stacks. The system will utilize a customized `init` script booting directly into the custom C++ `IntelligenceEngine` application, achieving a **sub-2 second boot-to-scan cold latency**.

## 2. The "All-American" Southern Sourcing Strategy

To align the product's marketing narrative with physical reality, we establish a **Southern Sourcing Strategy** to anchor yadiggg's assembly directly in the American South, leaning into the Atlanta/Georgia hip-hop roots (The Dungeon Family, Organized Noize, OutKast) that inspire the device.



### 2.1 Regional Manufacturing Partners

1. **Enclosure & Injection Molding: Cypress Industries** (Austin, TX). Cypress will handle the high-precision tooling and injection molding of the Smoke Amber polycarbonate top and bottom shells, ensuring a premium matte frosted finish that meets V0 flame retardant specifications.
2. **PCBA SMT Fabrication & Final Box-Build Assembly: Peach State Assembly / CM Partners** (Atlanta, GA). By routing high-speed surface-mount technology (SMT) component placement, manual mechanical integration (frame, display, battery),

and final quality control testing to a contract manufacturer in **Atlanta, Georgia**, we can legally and authentically print:

- **"Designed and Assembled in Atlanta, Georgia, USA"** on the physical aluminum skeleton frame.
- This physical stamp directly anchors the product's origin to the Atlanta roots of the sample-chopping hip-hop culture, providing an incredible branding narrative.

### 3. Logical Sequence of Decisions & Roadmap

|                 |              |                 |                  |
|-----------------|--------------|-----------------|------------------|
| +-----+         | +-----+      | +-----+         | +-----+          |
| PHASE 1         | PHASE 2      | PHASE 3         | PHASE 4          |
| OS & Software   | Electrical   | Mechanical &    | Factory RFQ      |
| EVK Prototyping | -->  PCBA V1 | -->  DFM Design | -->  & Pilot Run |
| (Weeks 1-6)     | (Weeks 7-12) | (Weeks 13-18)   | (Weeks 19-24)    |
| +-----+         | +-----+      | +-----+         | +-----+          |

#### Phase 1: Core OS & Software Stack Prototype (Weeks 1-6)

- **Milestone:** Run the complete software stack on off-the-shelf evaluation boards to prove software feasibility.
- **Key Decisions:**
  - Purchase an **i.MX 8M Nano Evaluation Kit (EVK)** and a **Sharp Memory LCD breakout board**.
  - Flash a custom Yocto Linux build to the EVK.
  - Benchmark OCR execution speed of **Tesseract-lite** and database lookup latency of **sqlite-vss** on the ARM A53 cores.
  - Port the **OLAF** acoustic fingerprinting C-library and verify local matching accuracy using a standard USB microphone.

#### Phase 2: Electrical Verification - PCBA V1 (Weeks 7-12)

- **Milestone:** Manufacture and verify the first custom 4-layer physical circuit board.
- **Key Decisions:**
  - Draft schematics routing the **Knowles MEMS microphones** directly to the native PDM inputs of the i.MX 8M Nano.
  - Route the **ESS ES9218PC DAC** strictly to the 3.5mm Aux jack for listening.
  - Incorporate the **Johanson Technology 2.45GHz Ceramic Chip Antenna** with its dedicated ground clearance zone.
  - Lay out the board to fit the target footprint (85mm x 50mm) and send files to **Advanced Circuits** (Aurora, CO) for quick-turn 4-layer fabrication.

#### Phase 3: Mechanical & Industrial Design (DFM) (Weeks 13-18)

- **Milestone:** Complete Design for Manufacturability (DFM) and resolve the physical depth envelope.
- **Key Decisions:**
  - **Enclosure Depth Decision:** Commit to the **Standard 15mm thickness** for the launch edition to maximize battery capacity (1500mAh) and lower PCB cost, or select the **Slim 12mm thickness** as a premium "Pocket Special" edition.
  - Verify the light-tight black-anodized aluminum isolation cylinder around the **AR1335 13MP sensor** via mechanical CAD to prevent light leakage through the translucent amber shell.
  - Perform thermal simulations to ensure the aluminum skeleton frame dissipates the heat of the Cortex-A53 cores during continuous scanning.
  - Order SLA rapid-prototyped translucent shells from **Protolabs** to test in-hand ergonomics.

#### Phase 4: Factory RFQ & Pilot Run (Weeks 19-24)

- **Milestone:** Quote final manufacturing tiers and run a 50-unit pilot assembly run.
- **Key Decisions:**
  - Send the completed Tech Pack, Gerber files, and mechanical STEP files to **Cypress Industries** (Austin, TX) for injection tool routing quotes.

- Send PCBA assembly files to **Peach State Assembly** (Atlanta, GA) for SMT and final box-build RFQ at the 500-unit MOQ tier.
- Run a 50-unit pilot assembly run in Atlanta to verify OCR accuracy, Sonic ID mic-isolation gaskets, and LRA haptic calibration.

## 4. "No-Book LLM" Prompts & Templates

To make this set of master documents highly effective and conversational when uploaded into ChatGPT Projects or a Custom GPT, use the following pre-engineered **System Prompts**.

### 4.1 Master Custom GPT System Prompt

Copy and paste this exact prompt into the **Instructions** box of your ChatGPT Custom GPT or Project:

You are the dedicated AI Product & Engineering Lead for "yadiggg" (formerly Crate IQ)—the ultimate offline physical companion device for vinyl crate diggers.

You have been uploaded with three master documents:

1. yadiggg\_product\_and\_interaction\_architecture.md
2. yadiggg\_hardware\_and\_electrical\_integration.md
3. yadiggg\_strategic\_decision\_map\_and\_roadmap.md

Your role is to act as an expert technical co-pilot. You must maintain complete consistency with the product's "Offline Soul" philosophy and its detailed hardware specifications.

When conversing with the team:

- Never suggest off-platform streaming or mobile cellular connectivity. If asked about "smart" app features, firmly reject them to preserve the device's purity and battery life.
- Keep hardware architecture technically accurate: Knowles digital MEMS microphones route directly via PDM to the i.MX 8M Nano SoC, the 3.5mm auxiliary port is strictly a playback output driven by the ESS ES9218PC DAC, and the Wi-Fi transceiver (Murata 1DX) requires a Johanson ceramic chip antenna with a dedicated ground clearance zone.
- Ground all user stories in the physical realities of vinyl digging (obscure basement record fairs, dusty sleeves, faded labels, and matrix run-out codes).
- Use these documents as your absolute "Source of Truth" (No-Book LLM mode). If a detail is not in the documents, make technically grounded inferences while explicitly stating your engineering assumptions.

### 4.2 Prompt Template to Generate Agile User Stories

Paste this prompt into the chat window to generate ready-to-use software development tickets:

Based on the "yadiggg\_product\_and\_interaction\_architecture.md" document, please generate 5 highly detailed Agile User Stories for the frontend and UI firmware development of the "Reveal Phase" on the 2.7" Sharp Memory LCD.

Each story must include:

1. User Story Description (As a... I want to... So that...)
2. Technical Acceptance Criteria (strictly utilizing the 400x240 1-bit monochrome resolution and 'Crate-Sans' 11px/16px/9px font specifications)

3. Haptic Feedback Trigger Conditions (LRA mechanical click simulation via DRV2605L)
4. Edge Case Handling (e.g., when the retrieved track metadata exceeds the screen length)

## 4.3 Prompt Template to Write Embedded C++ Drivers

Paste this prompt into the chat window to generate low-level firmware code:

Using the "yadiggg\_hardware\_and\_electrical\_integration.md" file, act as a Senior Firmware Engineer. Write a highly optimized, clean C++ class for the "IntelligenceEngine" capture sequence.

The class must:

1. Handle the physical button press interrupt (Orange Button) to wake up the system.
  2. Interface with the DW9714 VCM driver via I2C to trigger sub-100ms macro autofocus.
  3. Configure the MIPI CSI-2 receiver to stream a raw 13MP frame from the AR1335 image sensor into the memory buffer.
  4. Call the haptic LRA driver (DRV2605L) via I2C to trigger a sharp 15ms click effect once capture is confirmed.
- Include clear comments explaining the electrical registers and bus lines utilized.